Quantification: 877_wt_z081c1-2
==============================
PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	10
interior lacuna count	6
frame-edge lacuna count	4

Interior lacunae only: mean, median and sample SD (n = 6)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	3949.67	4538	2015.8	px^2
major_axis_length_px	lacuna shape	106.78	116.432	30.9776	px
minor_axis_length_px	lacuna shape	51.1454	47.3867	10.063	px
aspect_ratio	lacuna shape	2.084	2.0056	0.4995	unitless
eccentricity	lacuna shape	0.8509	0.8668	0.0899	unitless
solidity	lacuna shape	0.7779	0.782	0.1118	unitless
roots_count	headline	11	10	6.4807	count
ring_length_r30_px	headline	410.667	398.5	199.209	px
ring_length_r60_px	reported	1077.17	1078	408.818	px
owned_length_px	ownership-dependent	3125.75	2720.94	2012.05	px
edge_count	ownership-dependent	150	132.5	95.6284	count
mean_edge_length_px	ownership-dependent	21.2286	20.8698	1.8125	px
perimeter_px	normalised	338.183	339.304	102.716	px
ring_area_r30_px2	normalised	10595	11828	3909.13	px^2
ring_area_r60_px2	normalised	25259.3	28834.5	9190.38	px^2
in_frame_fraction_r30	normalised	1	1	0	unitless
in_frame_fraction_r60	normalised	1	1	0	unitless
ring_density_r30	normalised	0.0375936	0.036255	0.00600379	px^-1
ring_density_r60	normalised	0.0431012	0.0437166	0.00569186	px^-1
roots_per_100px_perimeter	normalised	3.0349	2.9399	1.1942	per 100 px
ring_attached_length_r30_px	network v2	356	316.5	211.627	px
ring_attached_length_r60_px	network v2	830	759	452.445	px
ring_length_w_r30_px	network v2	470.258	451.768	231.93	px
ring_length_w_r60_px	network v2	1242.71	1248.98	475.212	px
sholl_crossings_r10	network v2	9.3333	9.5	6.0222	count
sholl_crossings_r20	network v2	12.6667	15	4.5019	count
sholl_crossings_r30	network v2	16	16.5	5.4406	count
loop_count	loops, report only	13.3333	11	9.6678	count
loop_edge_share	loops, report only	0.3479	0.3548	0.1738	share
ring_width_mean_r30_px	width	7.1591	7.0847	0.5121	px

Per field
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measure	value	units
analysed_area_px2	1.0144e+06	px^2
total_skeleton_length_px	32376	px
canalicular_length_density_per_px	0.0319163	px^-1
junction_count	683	count
junction_density_per_px2	0.0006733	px^-2
lacuna_count	10	count

lacunae_per_px2	9.86e-06	px^-2
skeleton_component_count	329	count
mean_component_length_px	98.4073	px
median_component_length_px	25	px
field_length_density_w_per_px	0.036521	px^-1
field_density_without_flagged_per_px	0.0319163	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	491	px
unattached_component_count	29	count
canalicular_length_density_attached_per_px	0.0319163	px^-1
field_length_density_with_unattached_per_px	0.0324003	px^-1
loop_count	99	count
loop_edge_share	0.2492	share
edge_count	1645	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.4392	px
width_median_px	8	px
width_p10_px	4	px
width_p90_px	10.198	px
width_pixels_used	32376	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	828	47.9341	26.2785	1.8241	0.8363	0.811	1.1987	10.5652	544.853	True
2	4064	125.912	44.1728	2.8504	0.9364	0.8472	0.9481	38.4616	829.728	True
3	6500	135.386	68.4723	1.9772	0.8627	0.7101	0.7764	202.316	509.649	False
4	4498	128.704	46.6689	2.7578	0.9319	0.9623	0.3797	397.582	25.7919	True
5	4895	134.554	48.0863	2.7982	0.934	0.8829	0.4584	399.457	314.021	False
6	1388	60.0833	43.9829	1.3661	0.6813	0.6761	-0.1762	427.803	598.192	False
7	1690	77.7941	42.0098	1.8518	0.8417	0.6483	0.0159	450.566	636.056	False
8	5044	117.23	57.6341	2.034	0.8708	0.8539	1.376	449.833	814.625	False
9	1085	58.3256	24.5501	2.3758	0.9071	0.9313	0.4295	634.654	16.1935	True
10	4181	115.634	46.6872	2.4768	0.9149	0.8959	1.0406	724.949	827.982	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	8	225	625	1132.09	56	20.2158	7.3189
2	8	304	788	1030.69	45	22.9042	7.1897
3	20	720	1697	6827.89	321	21.2707	7.3764
4	6	247	711	1027.82	44	23.3596	8.1873
5	11	431	1176	2663.89	137	19.4444	8.0356
6	3	165	543	882.749	36	24.5208	6.5979
7	6	251	766	2108.3	103	20.4689	6.9792
8	17	531	1301	3493.66	175	19.9638	7.1902
9	0	65	236	0	0	0	7.9591

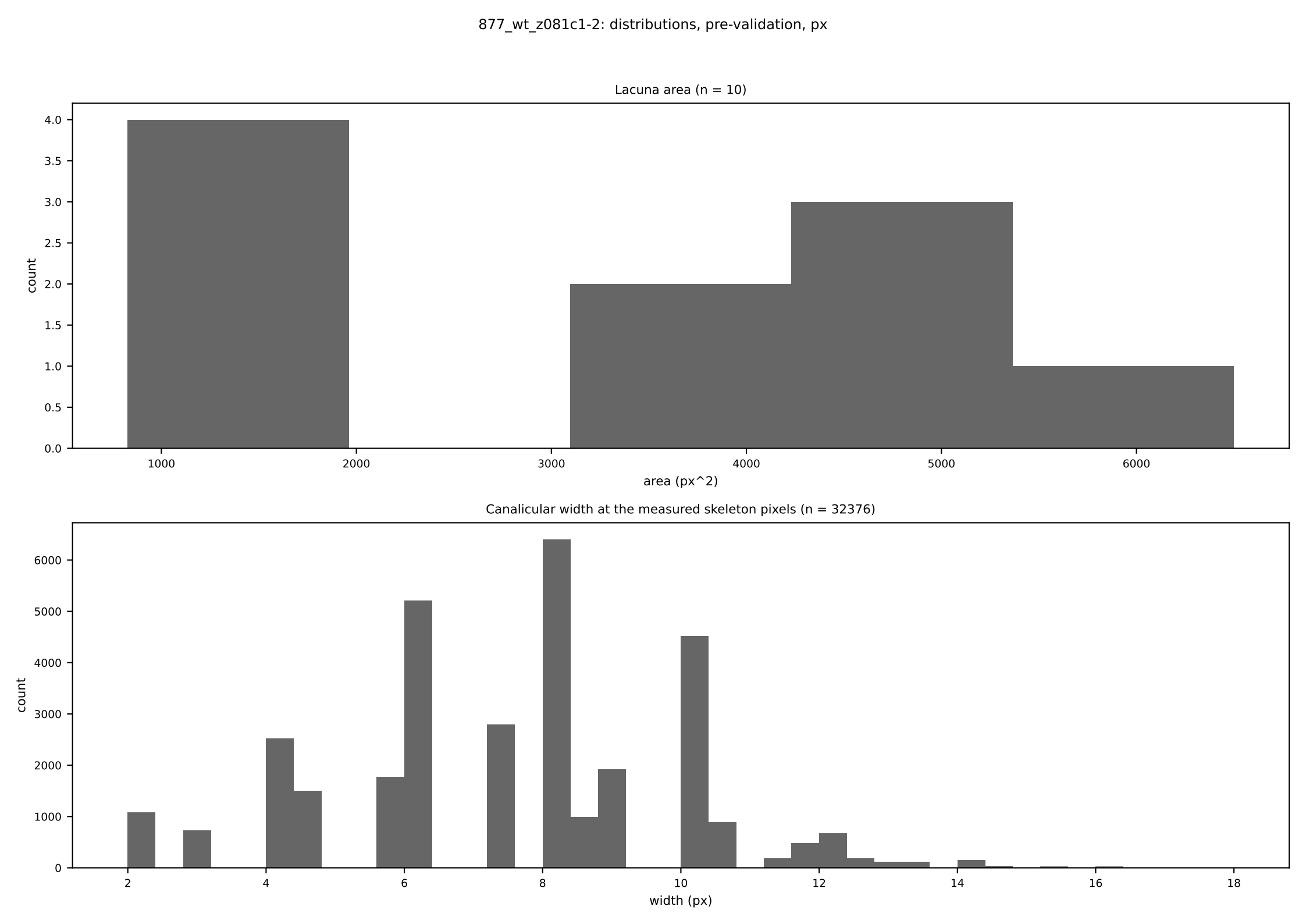
10 9 366 980 2777.99 128 21.703 6.7753

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	149.397	4144	11016	0.5976	0.5673	0.0542954	0.0567357	5.3549
2	351.404	9408	21117	0.7903	0.7218	0.0323129	0.0373159	2.2766
3	515.671	15609	35667	1	1	0.0461272	0.047579	3.8784
4	326.735	7997	18643	0.6576	0.6224	0.0308866	0.0381376	1.8364
5	354.617	12438	30402	1	1	0.0346519	0.0386817	3.1019
6	216.108	5237	12563	1	1	0.0315066	0.0432222	1.3882
7	263.451	6630	15255	1	1	0.0378582	0.050213	2.2775
8	355.262	12275	29427	1	1	0.0432587	0.0442111	4.7852
9	148.953	4761	12108	0.6788	0.6158	0.0136526	0.0194913	0
10	323.99	11381	28242	1	1	0.0321589	0.0347001	2.7779

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	225	571	264.693	727.34	6	7	7
2	276	705	348.563	910.021	9	8	10
3	710	1617	833.767	1970.95	19	17	25
4	205	584	278.966	816.896	6	9	10
5	356	926	487.517	1347.01	10	15	16
6	108	314	190.409	624.399	3	6	9
7	219	587	282.338	873.512	3	8	12
8	466	944	611.5	1489.44	12	15	17
9	0	0	74.9411	270.451	0	4	4
10	277	592	416.019	1150.94	9	15	17


877_wt_z081c1-2: distributions, pre-validation, px
Lacuna area (n = 10)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 32376)
count
width (px)